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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	MarketPulse – Stock Return Distribution & Risk Analytics System	
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Abstract

MarketPulse – Stock Return Distribution & Risk Analytics System is a Data Science-based analytical platform designed to study historical stock market behavior and evaluate investment risk using statistical analysis and data visualization. The system collects and processes historical stock price data to calculate daily returns, volatility, mean return, standard deviation, skewness, and kurtosis. It analyzes return distributions to identify trends, variations, and potential risk patterns.

The proposed system provides an interactive **Streamlit dashboard** that allows users to select stocks, analyze historical performance, visualize return distributions, compare multiple stocks, and identify periods of high volatility. Statistical and graphical techniques such as **histograms, box plots, correlation analysis, and moving averages** are used to present insights in an understandable manner. The system can also estimate the historical probability of positive and negative returns.

The project demonstrates the practical application of **Python, Pandas, NumPy, statistical analysis, and data visualization** in financial data science. Its main objective is to transform complex historical market data into meaningful insights that help users understand stock performance and associated risk. The system is intended as an **analytical and educational tool**, not as a guaranteed stock-price prediction or investment-advisory system.

Signature of the Guide HOD